

Logical operators:

1. && - logical and
2. || - logical or
3. ! – logical not

&&, || used to combine two or more expressions into a single expression.

! used for negation i.e. true becomes false and false become true.

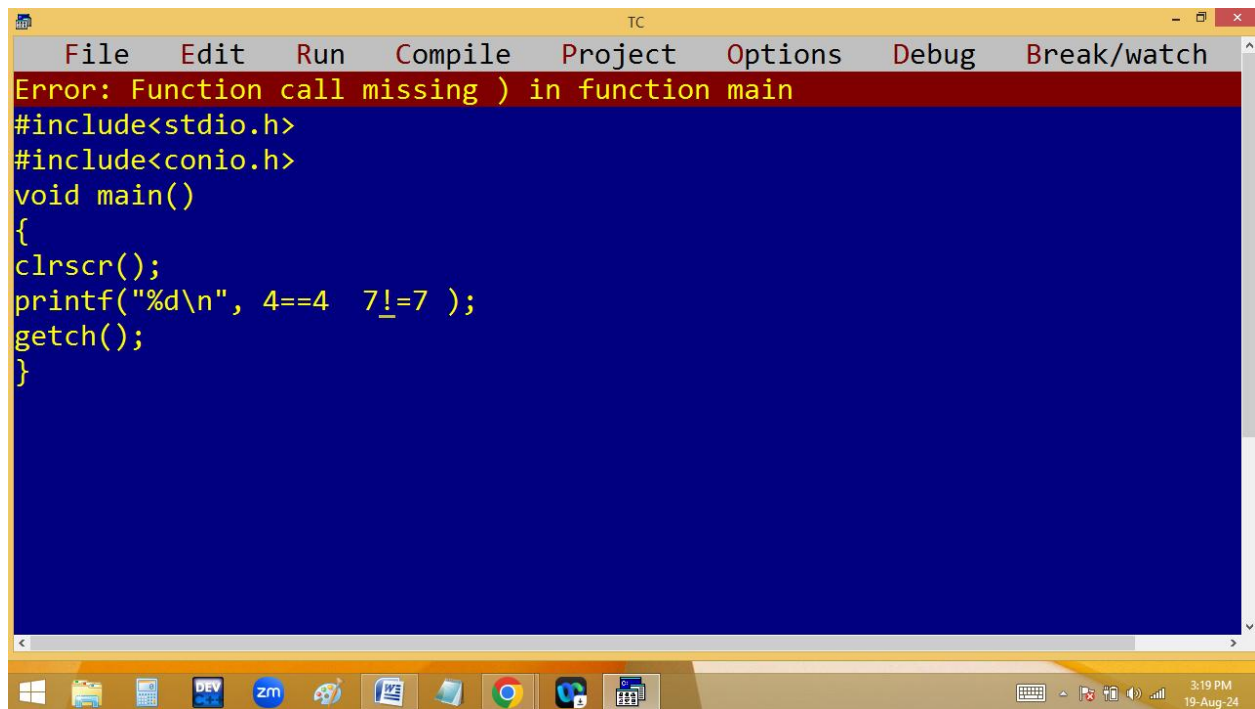
Note: In C && C++ other than 0 anything is 1 i.e. true.

truth tables:

Operator	Expression1	Expression2	Result
&& - and	1	1	1
	1	0	0
	0	1	0
	0	0	0
- or	1	1	1
	1	0	1
	0	1	1

!true = false

!false = true



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". A red error message banner at the top reads "Error: Function call missing) in function main". The code editor has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 4==4 7!=7 );
getch();
}
```

The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, DEV C++, Zoom, and other applications. The system clock in the bottom right corner shows "3:19 PM" and "19-Aug-24".

```
TC
Line 17 Col 29 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 4==4, 7!=7 );
printf("%d, %d\n", 4==4, 7!=7 );
printf("%d\n", 4==4 && 7!=7 );
printf("%d\n", 4==4 && 7<=7 );
printf("%d\n", 4!=4 || 7<=7 );
printf("%d\n", 4!=4 || 7!=7 );
printf("%d\n", !(4!=4));
printf("%d\n", 1==1 && 2<=2 && 3>=3);
printf("%d", 4==4 && 9<=9 || 7!=7 );
getch();
}
```

```
TC
1
1, 0
0
1
1
0
1
1
1
1_
```

1==1 && 2<=2

| && |

|

'a'>'b' || 'c'<='d'

97>98 || 99<=100

| || |

|

!(5==5)

|

!1

0

1==1 && 2<=2 && 3>=3

| && | && |

| && |

|

5==5 && 7!=8 || 8>9

| && | || |

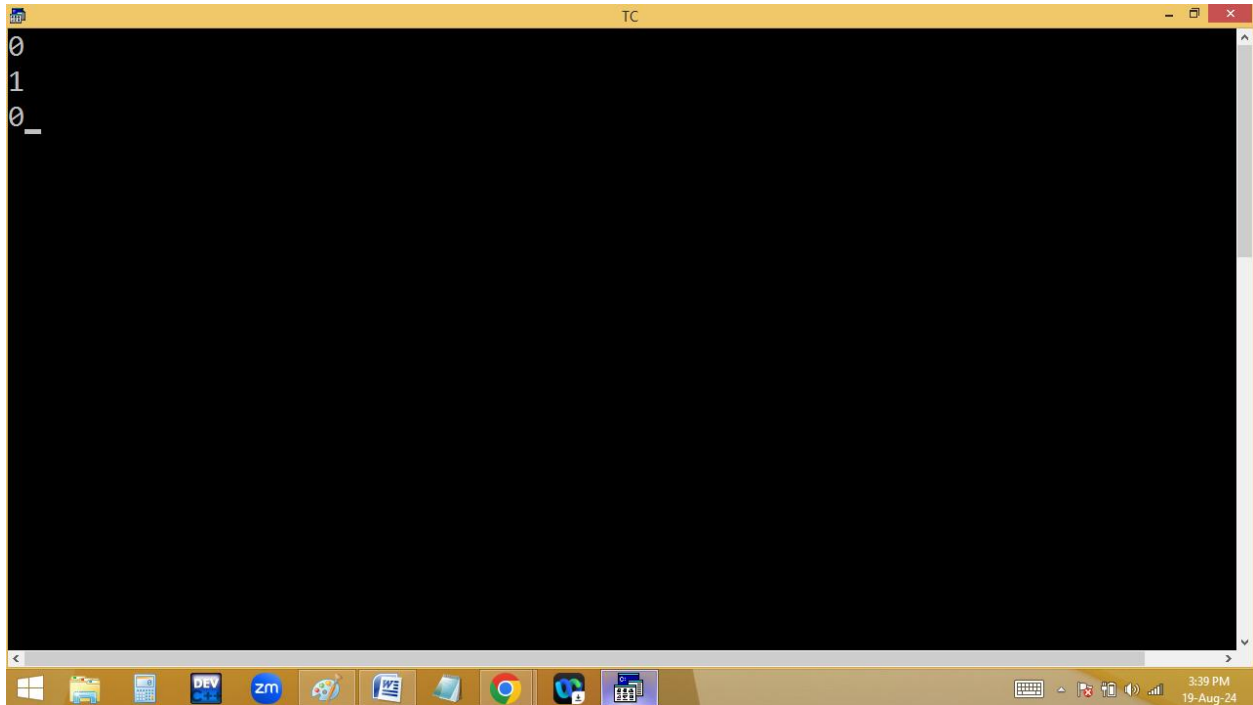
✓ | || 0

|

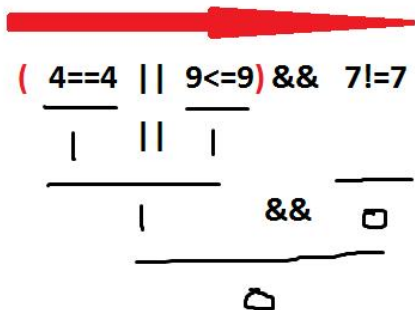
```

TC
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 4==4 && 9<=9 && 7!=7 );
printf("%d\n", 4==4 || 9<=9 && 7!=7 );
printf("%d", (4==4 || 9<=9) && 7!=7 );
getch();
}

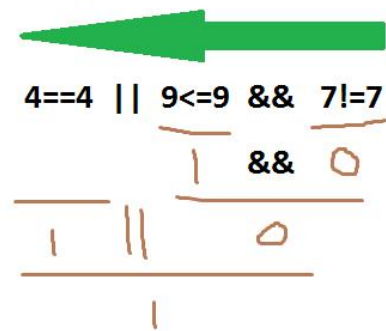
```



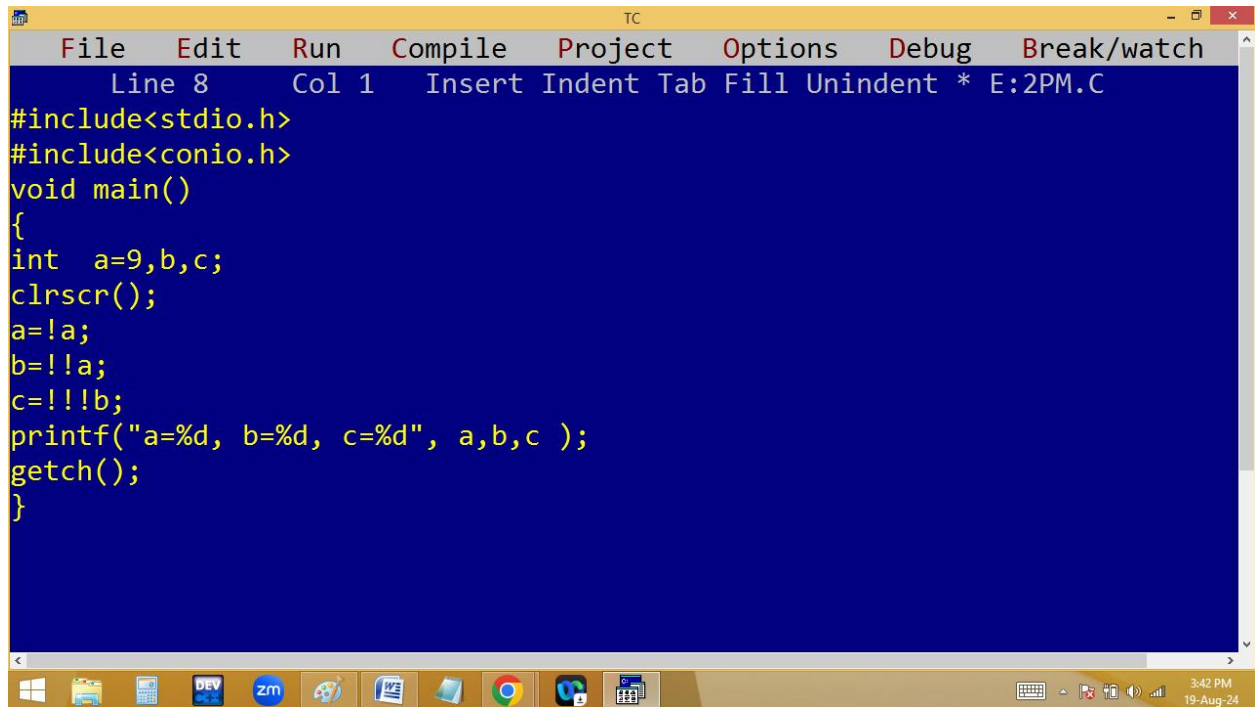
left association



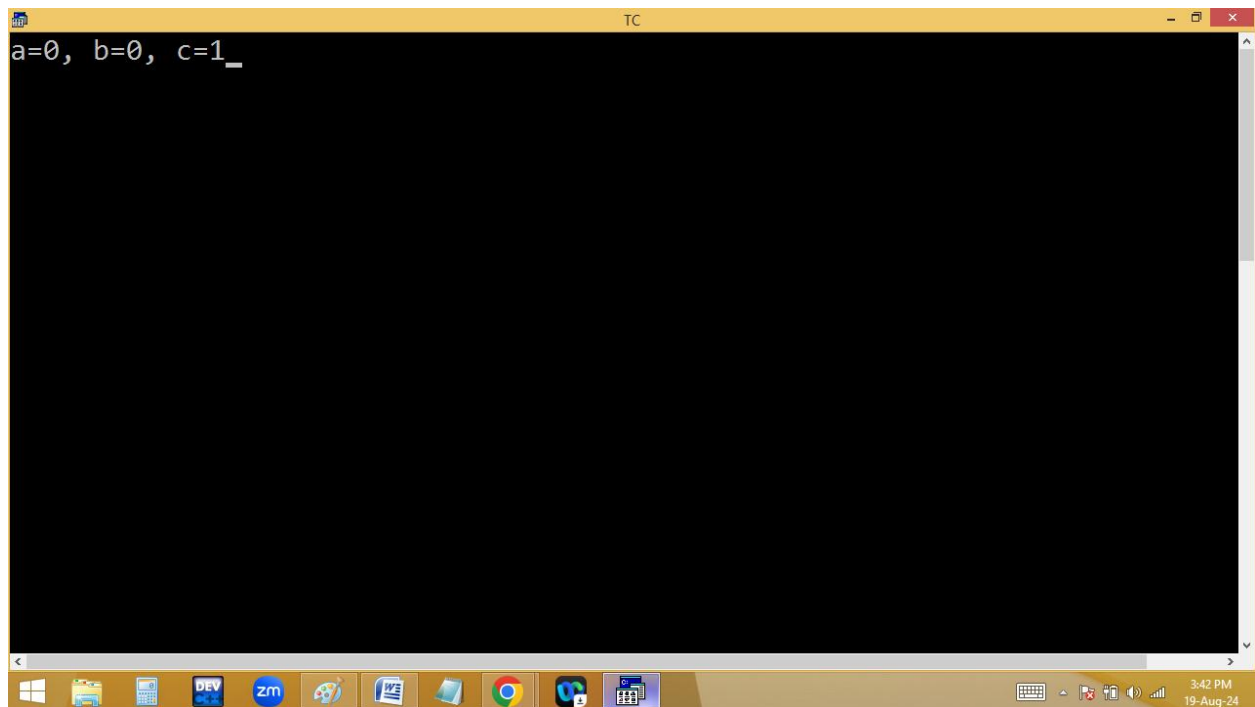
right association



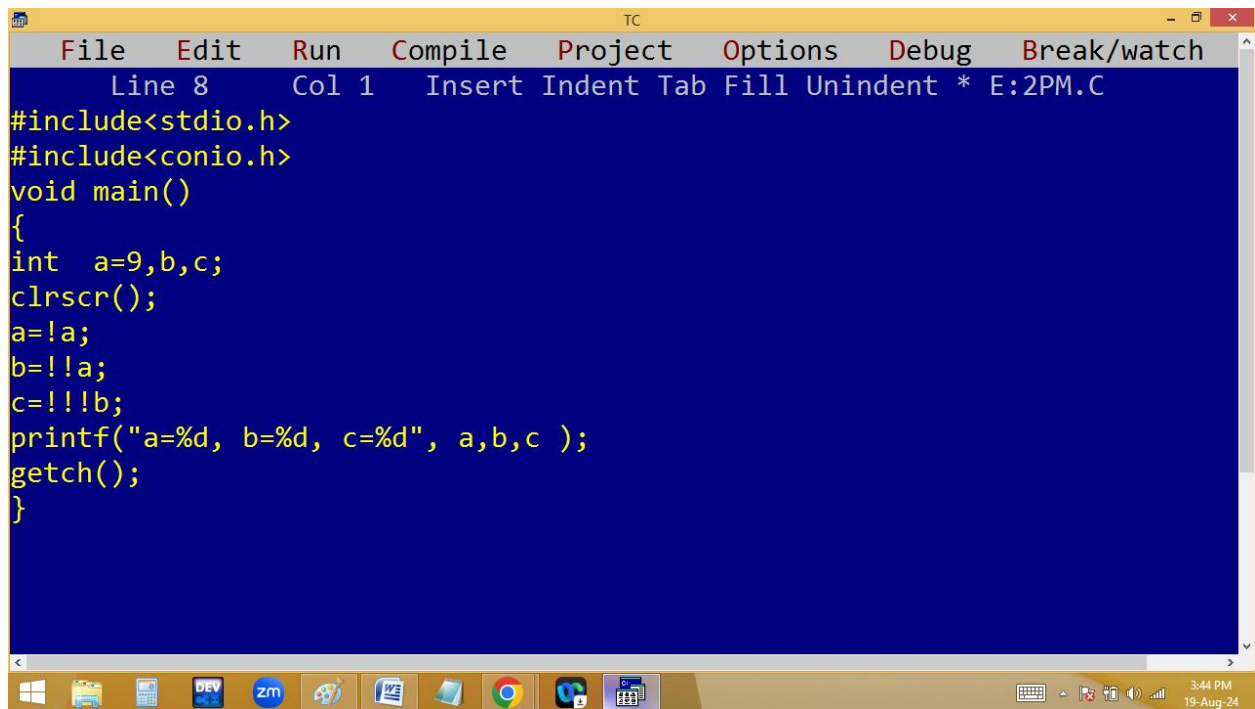
Note: When && , || operators are there, the first priority goes to &&.



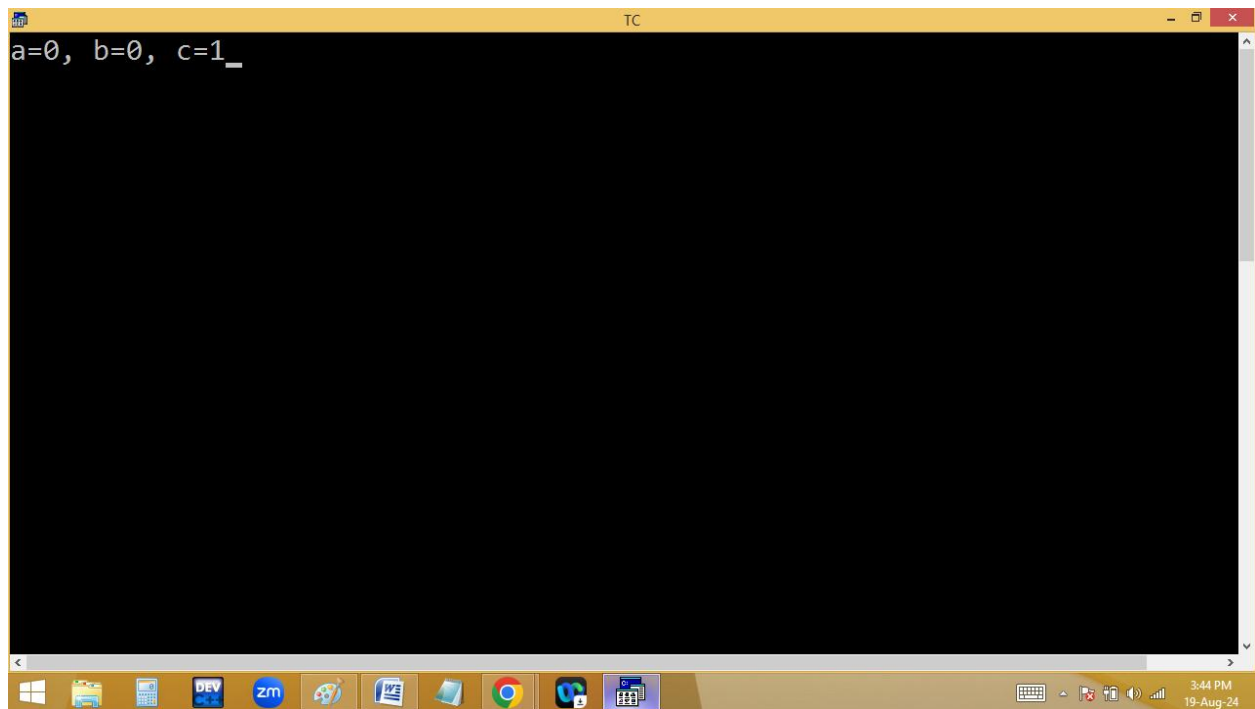
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=9,b,c;
clrscr();
a=!a;
b=!a;
c=!b;
printf("a=%d, b=%d, c=%d", a,b,c );
getch();
}
```



```
TC
a=0, b=0, c=1_
```



```
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=9,b,c;
clrscr();
a=!a;
b=!a;
c=!!!b;
printf("a=%d, b=%d, c=%d", a,b,c );
getch();
}
```

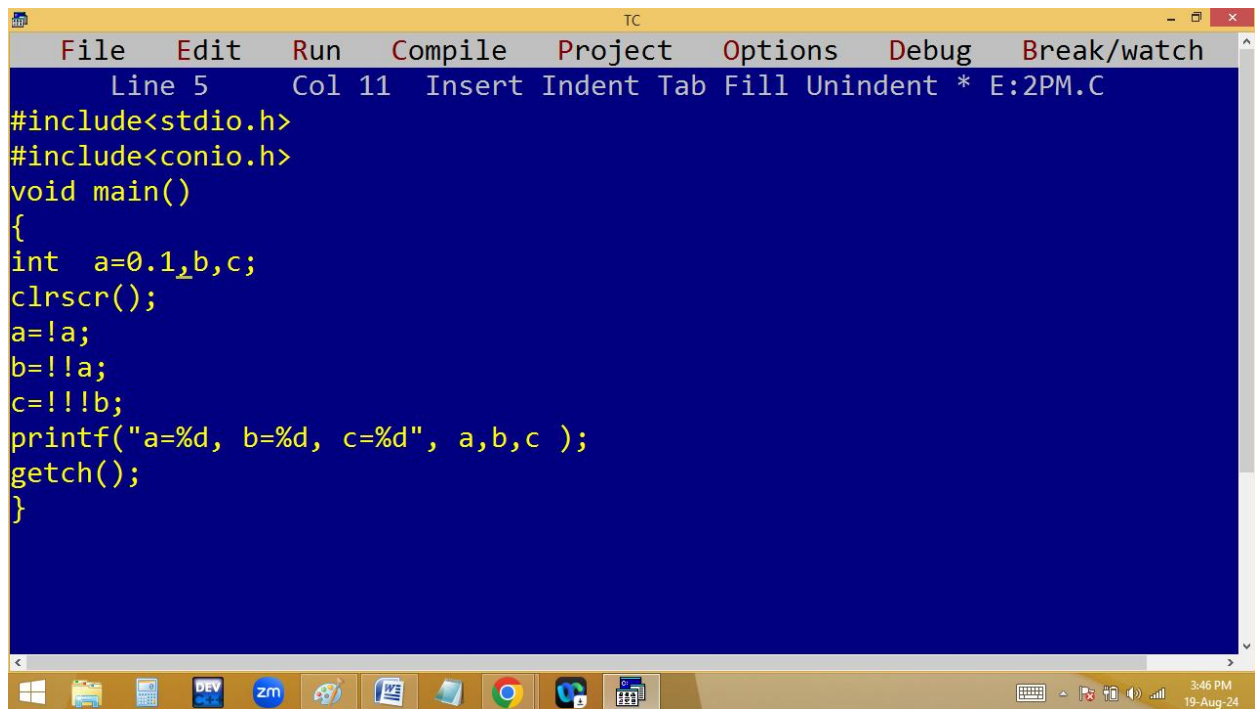


a=9

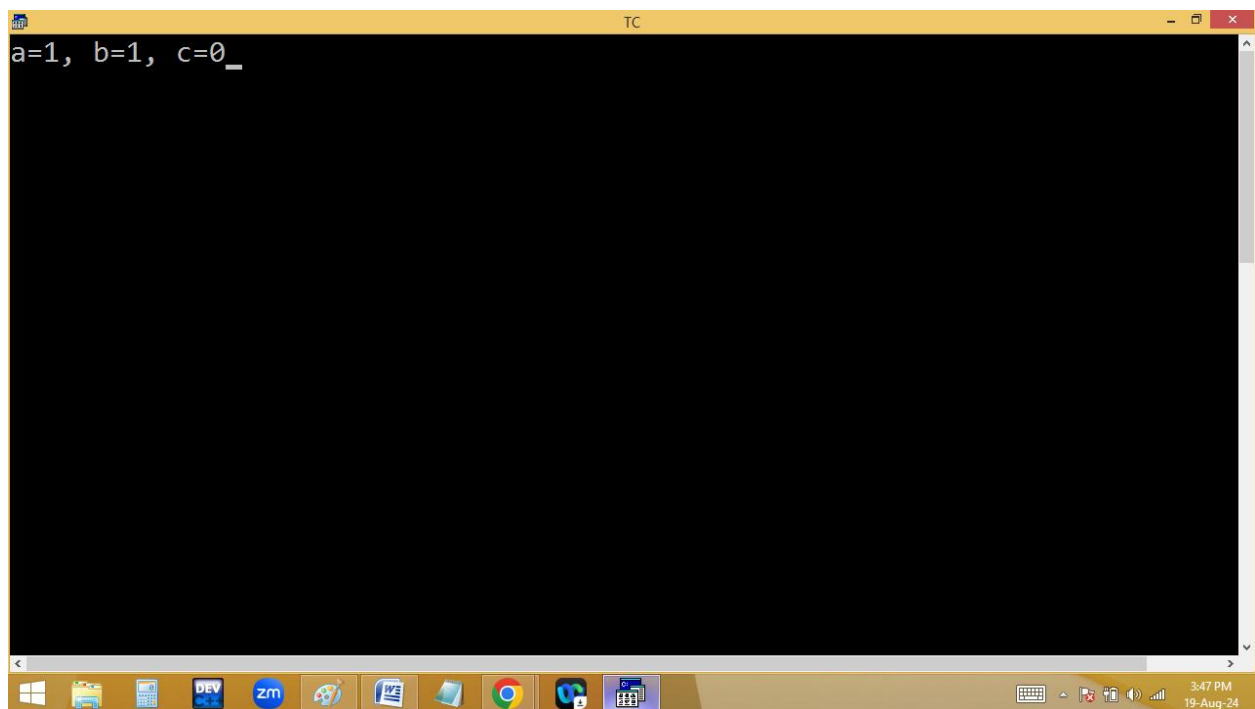
a=!a ==> a = !9 ==> a = !1 ==> a=0

b = !!a ==> b = !!0 ==> b=0

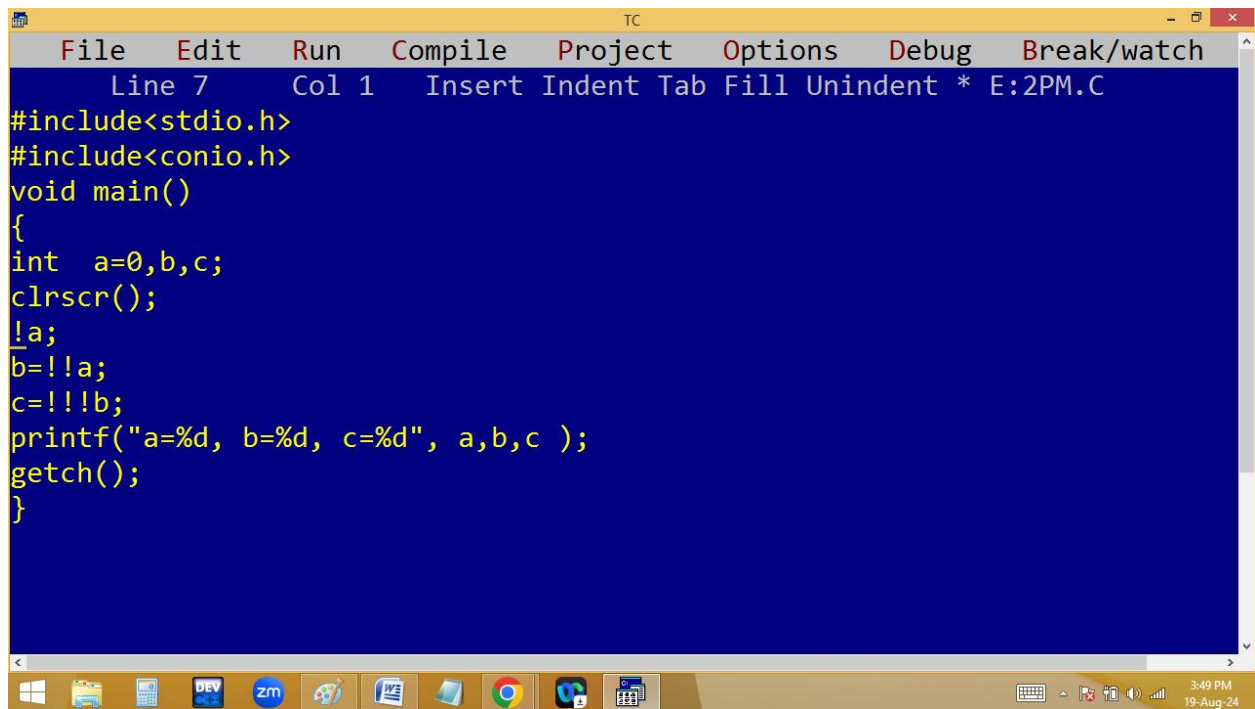
c = !!!b ==> c = !!!0 ==> c=1



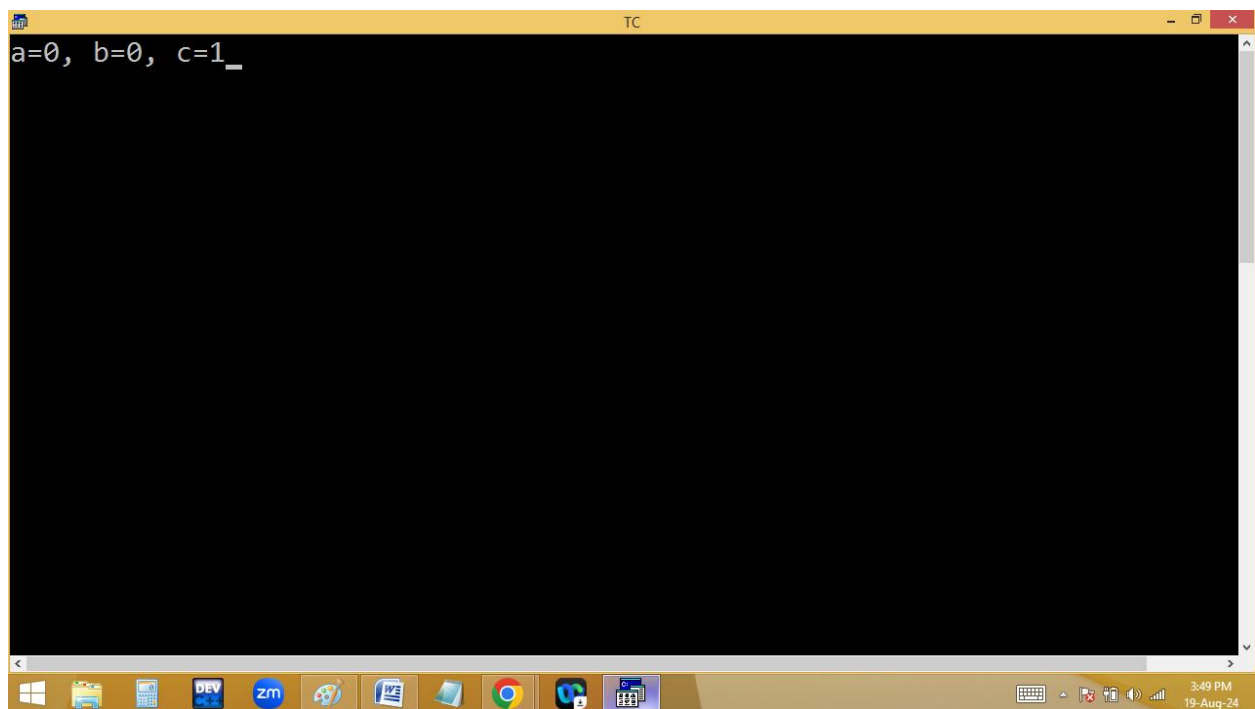
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 5 Col 11 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0.1,b,c;
clrscr();
a=!a;
b=!a;
c=!b;
printf("a=%d, b=%d, c=%d", a,b,c );
getch();
}
```



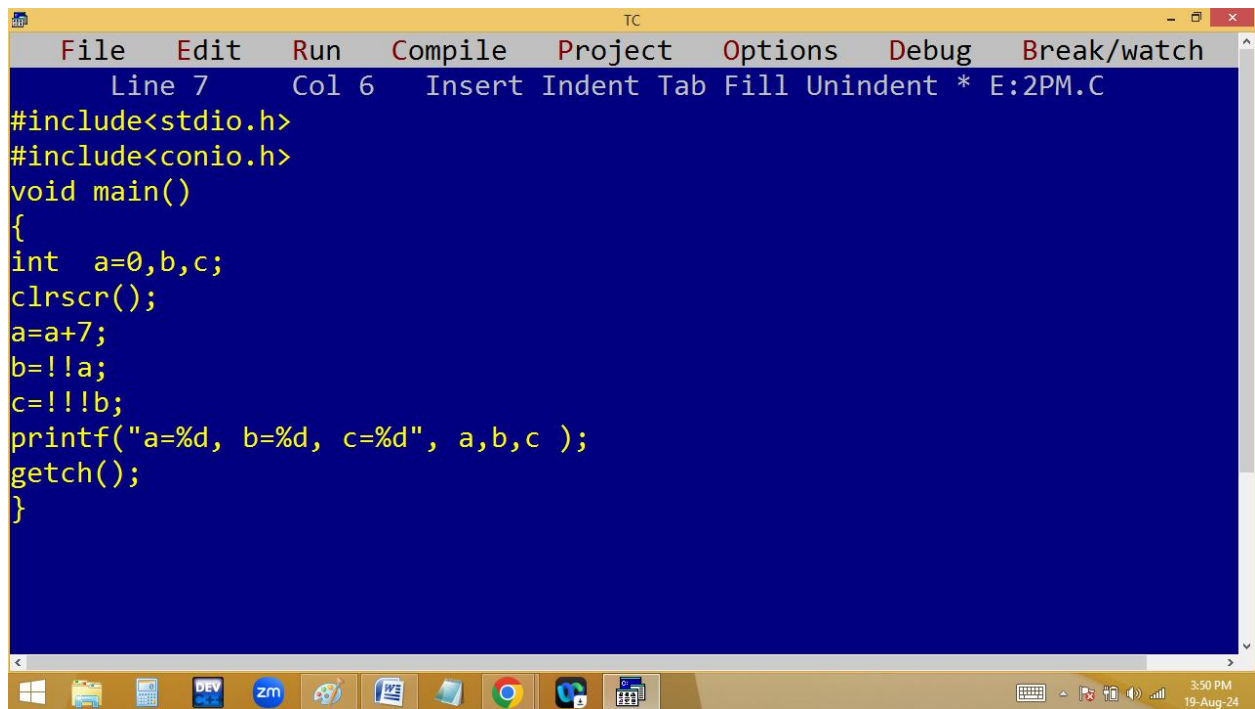
```
TC
a=1, b=1, c=0_
3:47 PM 19-Aug-24
```



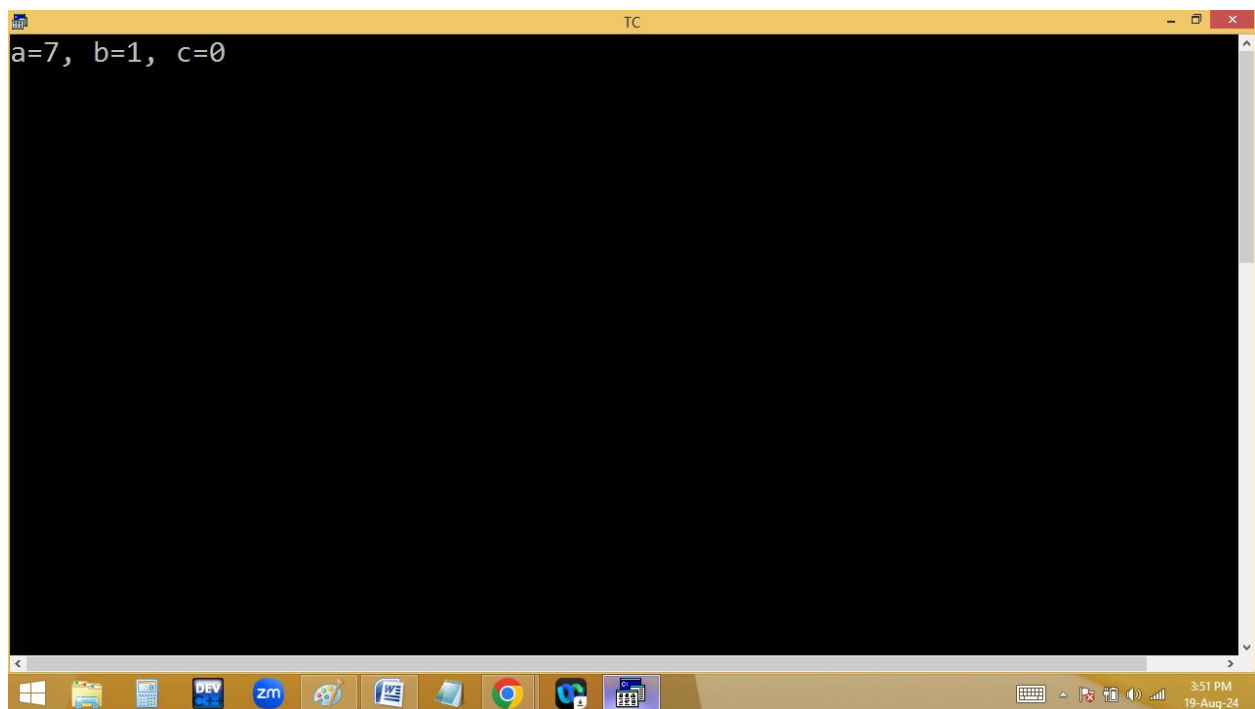
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0,b,c;
clrscr();
!a;
b=!!a;
c=!!!b;
printf("a=%d, b=%d, c=%d", a,b,c );
getch();
}
```



```
TC
a=0, b=0, c=1_
3:49 PM
19-Aug-24
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 6 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0,b,c;
clrscr();
a=a+7;
b=!!a;
c=!!!b;
printf("a=%d, b=%d, c=%d", a,b,c );
getch();
}
```



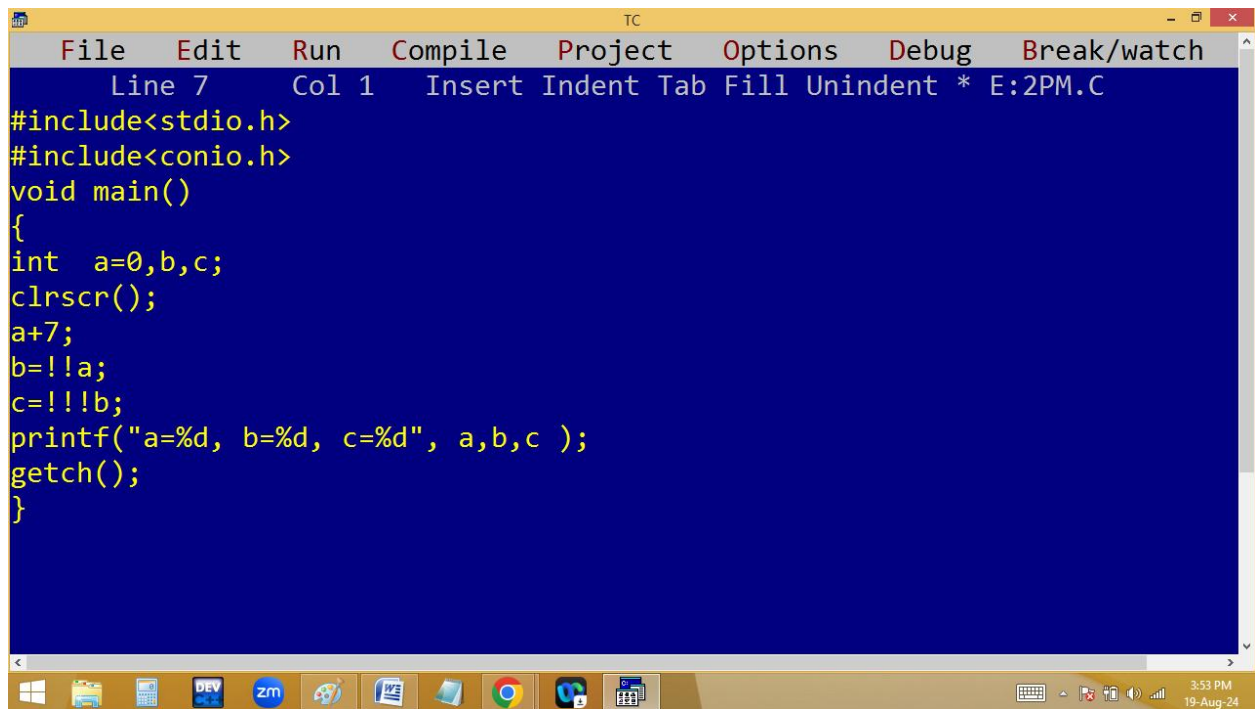
```
TC
a=7, b=1, c=0
```

a = 0

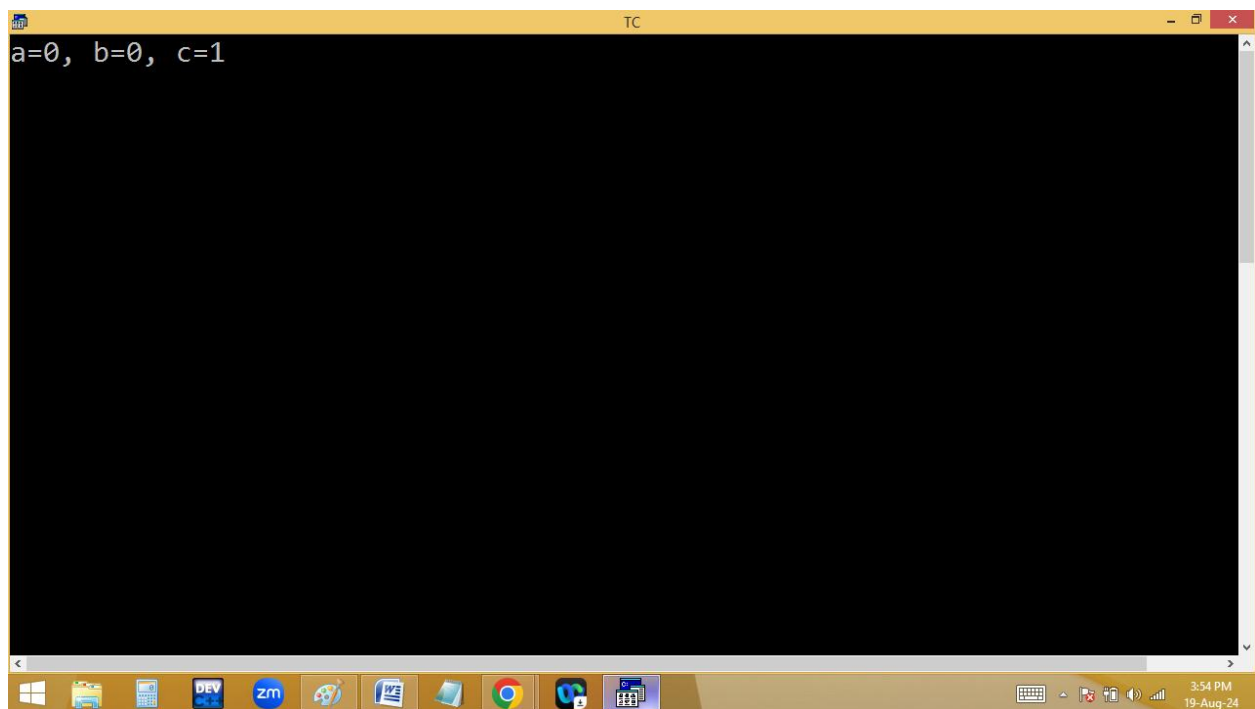
a=a+7 ==> 0+7 ==> a=7

b = !!a ==> !!7 ==> b=1

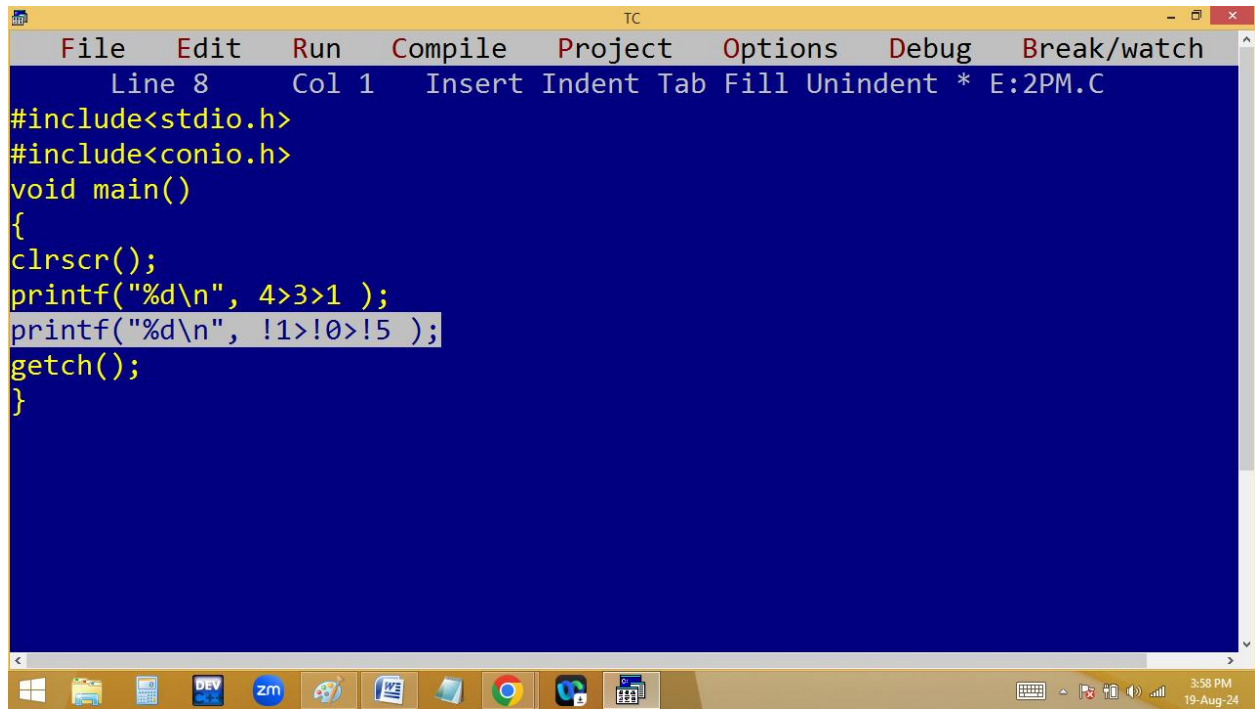
c = !!!b ==> !!!1 ==> c = 0



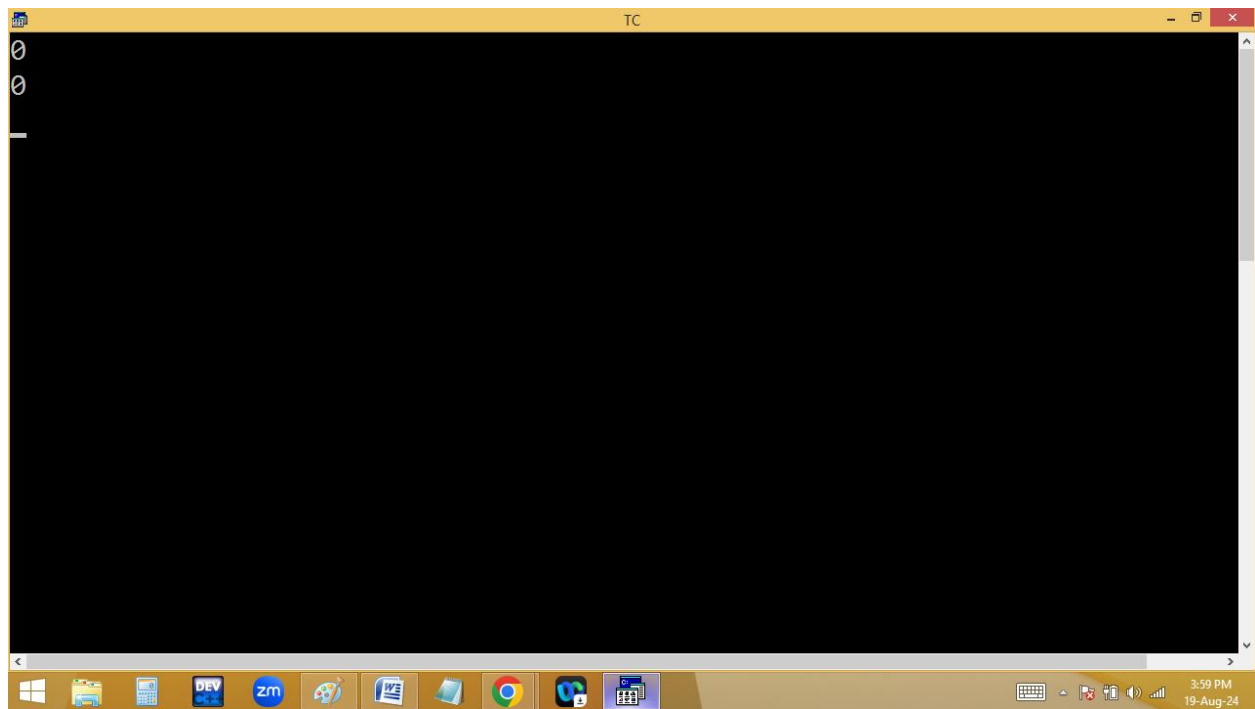
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0,b,c;
clrscr();
a+7;
b=!!a;
c=!!!b;
printf("a=%d, b=%d, c=%d", a,b,c );
getch();
}
```



```
TC
a=0, b=0, c=1
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 4>3>1 );
printf("%d\n", !1>!0>!5 );
getch();
}
```



$$\underline{\underline{4 > 3 > 1}}$$

$$\begin{array}{r} 1 > 1 \\ \hline 0 \end{array}$$